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Network Routing and Switching

CCE-415

THEORY ASSIGNMENT – 02

Comprehensive IPv6 Architecture, Protocols, and Practical Implementation

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1. Introduction to IPv6 and Motivation

IPv6 (Internet Protocol version 6) was developed mainly to overcome the limitations of IPv4, especially the shortage of available IP addresses. It provides a much larger address space and several improvements for modern networks.

- **Large Address Space:** IPv6 uses a 128-bit address, providing approximately 3.4×10^{38} possible addresses. This is far greater than the 32-bit address space of IPv4.
- **Reduced Dependence on NAT:** The huge number of IPv6 addresses makes it possible to assign globally unique addresses to a large number of devices without relying heavily on Network Address Translation (NAT).
- **Simplified Header:** IPv6 uses a fixed 40-byte base header, which makes packet processing more efficient than the variable-length IPv4 header.
- **Automatic Configuration:** IPv6 supports Stateless Address Autoconfiguration (SLAAC), allowing hosts to configure their addresses automatically using information provided by routers.
- **Improved Security Support:** IPv6 was designed with native support for IPsec, which can be used to provide authentication and encryption at the network layer.

2. IPv6 Address Architecture and Representation Rules

An IPv6 address is 128 bits long and is normally written as eight groups of four hexadecimal digits. Each group is separated by a colon.

For example:

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

IPv6 provides rules for shortening long addresses.

1. Removing Leading Zeros:

Leading zeros in any hexadecimal group can be omitted.

For example:

```
0db8 -> db8
0042 -> 42
```

Therefore,

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

can be shortened by removing leading zeros.

2. Using Double Colon (::):

One continuous sequence of zero groups can be replaced by `::`. This compression can be used only once in an IPv6 address.

For example:

```
2001:db8:0000:0000:0000:ff00:42:8329
```

can be written as:

```
2001:db8::ff00:42:8329/64
```

3. Prefix Length:

The prefix length is written after a slash and specifies how many bits represent the network portion of the address. A `/64` prefix is commonly used for IPv6 end-user network segments.

3. IPv6 Address Types and Scopes

IPv6 supports several address types, each designed for a specific communication purpose.

Address Type	Prefix	Scope	Main Use
Global Unicast (GUA)	2000::/3	Global / Internet	Publicly routable communication between hosts and networks.
Unique Local (ULA)	fd00::/8	Local / Organization	Private communication inside an organization or internal network.
Link-Local (LLA)	fe80::/10	Link / Segment	Local communication, NDP, and next-hop operations.
Multicast	ff00::/8	Variable	One-to-many communication and group-based services.
Anycast	Same as unicast	Nearest Node	Allows communication with the nearest member of a group of nodes using the same address.

Unlike IPv4, IPv6 does not use broadcast addresses. Multicast is used instead for many-to-many or one-to-many communication requirements.

4. Configuration Mechanisms: SLAAC, EUI-64, and DHCPv6

IPv6 hosts can obtain addresses and other network information through several mechanisms, including SLAAC, EUI-64, and DHCPv6.

SLAAC (Stateless Address Autoconfiguration):

SLAAC allows a host to automatically configure an IPv6 address without requiring a DHCPv6 server for the address itself.

- The router sends Router Advertisement (RA) messages containing the network prefix.
- The host normally uses a /64 prefix to construct its address.
- The remaining 64 bits are generated automatically, using a method such as EUI-64 or a randomized interface identifier.

EUI-64 Address Generation:

An IPv6 interface identifier can be generated from a 48-bit MAC address using EUI-64.

For example, consider the MAC address:

00:11:22:33:44:55

The process is:

1. Divide the MAC address into two halves.
2. Insert `ff:fe` between the two halves.
3. Invert the Universal/Local (U/L) bit of the first byte.

The resulting interface identifier becomes:

0211:22ff:fe33:4455

DHCPv6:

DHCPv6 can provide additional configuration information to IPv6 hosts.

- **Stateful DHCPv6:** The DHCPv6 server assigns IPv6 addresses and can also provide other configuration parameters such as DNS information.
- **Stateless DHCPv6:** The host obtains its IPv6 address through SLAAC, while DHCPv6 provides additional information such as DNS servers and domain names.

5. Key Protocols: NDP and ICMPv6

Neighbor Discovery Protocol (NDP):

NDP is an important IPv6 protocol that works through ICMPv6. It performs functions that were handled by ARP, ICMP Router Discovery, and ICMP Redirect in IPv4.

Important NDP messages include:

- **Neighbor Solicitation (NS):** Used to discover the link-layer address of a neighboring device and for Duplicate Address Detection (DAD).
- **Neighbor Advertisement (NA):** Sent as a response to Neighbor Solicitation and provides information about the neighboring device.
- **Router Solicitation (RS):** A host sends an RS message to request router information.
- **Router Advertisement (RA):** A router sends RA messages containing network prefixes, configuration information, and default-router information.

ICMPv6:

ICMPv6 provides several important functions in IPv6 networks. It is used for error reporting, network diagnostics, Path MTU Discovery, and communication related to multicast group management.

For example, IPv6 connectivity can be tested using tools such as:

```
ping6
```

or, on systems that support the newer syntax:

```
ping -6
```

6. Subnetting and Address Aggregation

In IPv6 networks, a /64 prefix is commonly used for an individual LAN or end-user network. This leaves 64 bits for interface identification.

Suppose an organization receives the following IPv6 allocation from an ISP:

```
2001:db8:1000::/48
```

To create /64 subnets:

$$64 - 48 = 16 \text{subnetbits}$$

Therefore, the number of available /64 subnets is:

$$2^{16} = 65,536$$

For example, different departments can be assigned separate subnets:

- **Engineering:** 2001:db8:1000:0001::/64
- **Sales:** 2001:db8:1000:0002::/64
- **Management:** 2001:db8:1000:0003::/64

Address Aggregation:

Instead of advertising every individual /64 subnet to upstream routers, the organization can advertise the larger /48 prefix:

2001:db8:1000::/48

This route summarization reduces the number of entries that need to be maintained in larger routing tables.

7. Transition Mechanisms and Security

IPv6 transition mechanisms are used to allow IPv6 networks to communicate during the period when IPv4 and IPv6 coexist.

1. Dual Stack:

In a dual-stack network, devices run IPv4 and IPv6 simultaneously. This allows communication with both IPv4-only and IPv6-enabled systems.

2. Tunneling:

Tunneling can carry IPv6 traffic through an IPv4-based network by encapsulating IPv6 packets inside IPv4 packets. Examples include techniques such as 6to4 and GRE tunneling.

3. Translation:

Technologies such as NAT64 and DNS64 allow IPv6-only clients to communicate with IPv4-only servers by translating between the two protocols.

Security and Privacy:

- **IPsec:** IPv6 supports IPsec for providing authentication, integrity, and encryption at the network layer.

- **Privacy Extensions:** IPv6 privacy extensions can generate temporary randomized interface identifiers. This makes it more difficult to continuously track a device based only on its IPv6 interface identifier.

8. Practical Implementations and Troubleshooting

1. Basic Linux IPv6 Configuration and Verification:

An IPv6 address can be manually assigned to a Linux interface using the following command:

```
sudo ip -6 addr add 2001:db8:1:1::10/64 dev eth0
```

Connectivity can then be tested with:

```
ping6 -c 4 2001:db8:1:1::1
```

The IPv6 path can be examined using:

```
traceroute6 2001:db8:1:1::1
```

2. DHCPv6 Server Configuration:

A basic DHCPv6 configuration can be defined in `/etc/dhcp/dhcpd6.conf`:

```
subnet6 2001:db8:1:2::/64 {  
    range6 2001:db8:1:2::100 2001:db8:1:2::200;  
    option dhcp6.name-servers 2001:db8:1:2::1;  
    option dhcp6.domain-search "pstu.ac.bd";  
}
```

3. SLAAC Troubleshooting Case Study:

Problem:

Suppose hosts are unable to automatically configure IPv6 addresses through SLAAC.

Diagnosis:

The Router Advertisement (RA) message should be checked carefully. Two important flags are:

- **Managed Address Configuration (M-flag):** If this flag is set to 1, hosts are instructed to obtain address configuration through DHCPv6.
- **Autonomous Address Configuration (A-flag):** If the A-flag is set to 0 for the advertised prefix, hosts cannot use that prefix for automatic SLAAC address generation.

Solution:

For normal SLAAC operation, the router should advertise a suitable prefix with the **A-flag set to 1** and appropriate prefix lifetimes. The router advertisements and interface configuration should then be verified again.